AI for Prediction of Performance of Metal Hydrides for Hydrogen

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Abstract. Green Hydrogen has emerged as a promising candidate for facilitating the transition towards cleaner energy sources. However, the efficient storage of hydrogen remains a significant challenge in establishing it as a prominent clean fuel. Among the various techniques employed for hydrogen storage, metal hydrides have gained widespread attention due to their distinct advantages over alternative methods due to their superior safety and higher gravimetric and volumetric storage capacities. To gain insight into the performance of metal hydrides during their operation, different machine-learning techniques were used for the prediction of performance. A dataset was constructed using findings from a secondary source, in conjunction with the open-source Materials Project database. The dataset includes 11 features and a dependent variable, which is the discharge capacity of the La-Ni-Mg Alloy. Data was collected after the alloy had been used for 100 cycles and was visualized using the open-source software, Orange. Four regression models - Support Vector Regression, Random Forest, Decision Tree, and K Nearest Neighbour (KNN) - were employed. The predicted values from these supervised models were compared with the actual values. The evaluation matrix indicated that the Support Vector Regression model performed the best, with a adjusted R-Squared value of 0.9619. The findings of this study could contribute to the development and optimization of metal hydrides as a viable alternative for hydrogen storage. Furthermore, the proposed methodology could be extended to other types of materials and systems for hydrogen storage.

Keywords. Green Hydrogen Storage, Metal Hydrides, Machine Learning, Performance Analysis

1. **Introduction**

The global shift towards sustainable energy has propelled green hydrogen to the forefront of clean energy solutions, promising efficient energy storage and emission-free power generation [1]. Hydrogen holds great promise as a clean energy source for the future. It’s role in energy transition is bound to expand as the research and development on various aspect of its use, storage and distribution

continues. Hydrogen storage among all remains a key technology for the advancement of hydrogen and fuel cell technologies. Although hydrogen has the highest energy per mass of any fuel, its low density in ambient temperatures results in a low energy per unit volume[2]. Hydrogen can be stored in gaseous, liquid, or solid states. Storing hydrogen as a gas necessitates high-pressure tanks, while liquid storage requires cryogenic temperatures[2].Key to this transformation are metal hydrides, offering the potential to securely store and release hydrogen at ambient temperatures [3]. Yet, a substantial barrier to their
widespread use remains: the challenge of metal hydride degradation [4].This degradation not only compromises the integrity of hydrogen storage systems but also poses a barrier to the viability of hydrogen as a cornerstone of future energy systems.

In response to these challenges, this paper explores the integration of advanced artificial intelligence (AI) and machine learning techniques to address the intricate issue of metal hydride degradation. Leveraging the computational prowess of AI, alongside a wealth of experimental and simulation data, this study aims to develop predictive models that can identify and anticipate degradation patterns. These models, adept at analysing complex datasets, hold the promise of enhancing the stability and performance of metal hydrides, thereby supporting the broader adoption of hydrogen-based energy solutions.

Metal hydrides are a class of solid-state materials that can absorb and release hydrogen. With adequate data on the different parameters of metal hydrides based on its operation, the discharge capacity of the metal hydrides can be predicted using Machine Learning. Machine Learning is the ability of computers to learn and extract patterns without being explicitly being programmed. Based on the data available different Machine Learning techniques can be used for the predicting a feature of system in the future. For this paper such a dataset was obtained from a research based on La–Mg-Ni-based alloys done by Liu et.al[5]. The dataset includes the 12 different features including the discharge capacity of the metal hydrides on 100 cycles of charging and discharging the La_2MgNi_9 alloy. This paper will delve into the intricacies of machine learning model development, from feature formulation to model selection, and demonstrate their utility in predicting metal hydride degradation.

2. Materials and methods

The overall schematic diagram of the study is shown in figure 1.

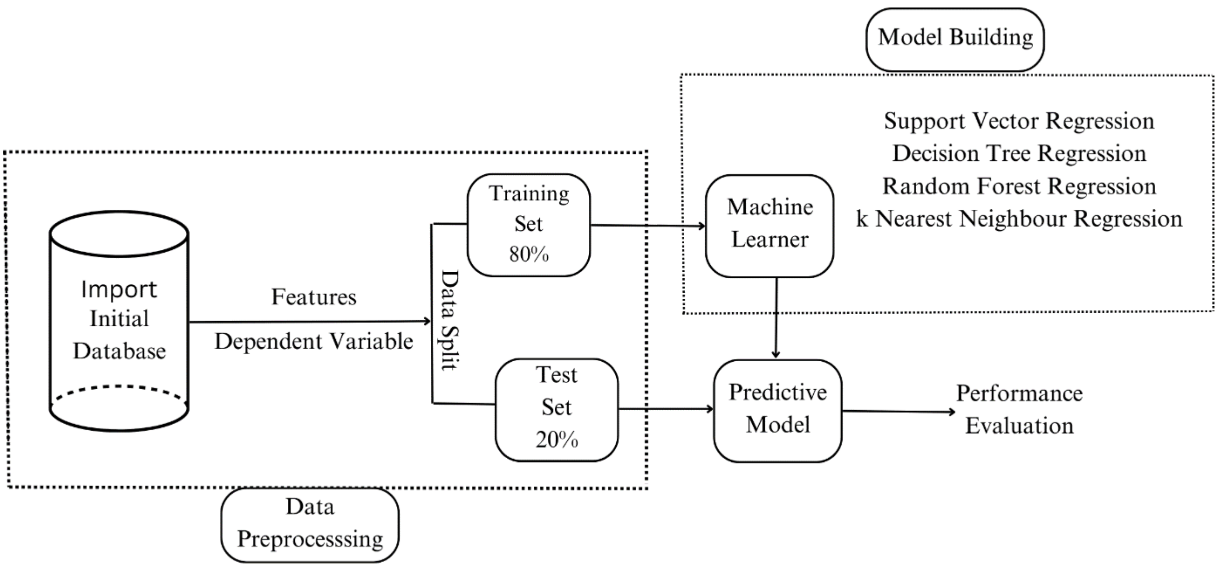

Figure 1: Schematic diagram of the study

2.1 Study Material

La-Mg-Ni based hydrogen storage alloys exhibits promising electrochemical properties, including large discharge capacity, easy activation and good high-rate discharge capacity[6] but it has a limited application is due to their fast degradation in alkaline electrolyte[7]. In this research, Li_2MgNi_9 made by mixing the precursor powders of $LaMgNi_4$ and $LaNi_5$ in the molar ratio of 1:1 is taken as the study material. The material has properties shown in table 1 in regards to hydrogen storage.

Table 1: Hydrogen Storage Properties of Li_2MgNi_9

Properties	Value
Hydrogen content during absorption (wt. %)	0 - 1.1 (at 303 K)
Hydrogen content during desorption (wt. %)	0.2 – 2.2 (at 303 K)
Discharge Capacity	Maximum = 390 mAh g^{-1}
	Minimum = 250 mAh g^{-1}
Strain (%) at charging	Maximum = 0.75
	Minimum = 0.1
Strain (%) at discharging	Maximum = 1
	Minimum = 0.1

2.1 Dataset Preparation

In this research, datasets were derived from secondary sources, which presented data in graphical form, making it necessary to extract numerical values from the graphs. In this study, the process of constructing a dataset from graph-based information was explored using an open-access software tool called Web Plot Digitizer. To begin, a screenshot of the graph of interest was obtained. This screenshot was imported into the Web Plot Digitizer[8] interface. The axes of the graph in the software were aligned with the original axis of the screenshot, ensuring accurate coincidence of both the x-axis (horizontal) and y-axis (vertical). Values along each axis were manually inputted, specifying whether they followed a linear or logarithmic scale. Each data point on the graph was carefully tracked, clicking on the corresponding location in the Web Plot Digitizer interface. Once all data points were tracked, the software provided numerical coordinates for each point, representing the x and y values of the data. These values were compiled to create the dataset. The use of Web Plot Digitizer streamlined the process of extracting data from graphical representations, allowing researchers to confidently construct datasets from secondary sources for further analysis.

2.2 Dataset Visualization

A Free Viz plot was constructed as shown in figure 1 to see relationship of independent variables(features) which are 11 in number with respect to the target (dependent variable) in a 2-D plane. The general explanation during the visualization is listed

- Pressure: Higher pressures might allow for more hydrogen to be stored, increasing the discharge capacity.
- Hydrogen Content in Absorption (%)**:** A higher absorption percentage could mean more hydrogen is available for discharge, thus increasing the discharge capacity.

- Temperature (K)**:** At higher temperatures, the discharge capacity might decrease due to increased kinetic energy of the hydrogen atoms.
- Hydrogen Capacity in Desorption (%): A higher hydrogen capacity in desorption lead to a higher discharge capacity.
- Cycles: The discharge capacity decrease over many cycles due to wear and tear on the storage system.
- Particle Diameter (nm): The size of the particles in the storage material affects the surface area available for hydrogen absorption, impacting the discharge capacity.
- Differential Volume (%)**:** This refers to changes in the volume of the storage system or material. Large volume changes could affect the discharge capacity.
- Charge/Discharge Depth (%)**:** A deeper depth of discharge lead to a higher discharge capacity but also lead to faster degradation of the system.
- Strain Rate in Charging**:** A high strain rate decrease the discharge capacity by damaging the material.
- Strain Rate in Discharging: A high strain rate decrease the discharge capacity.

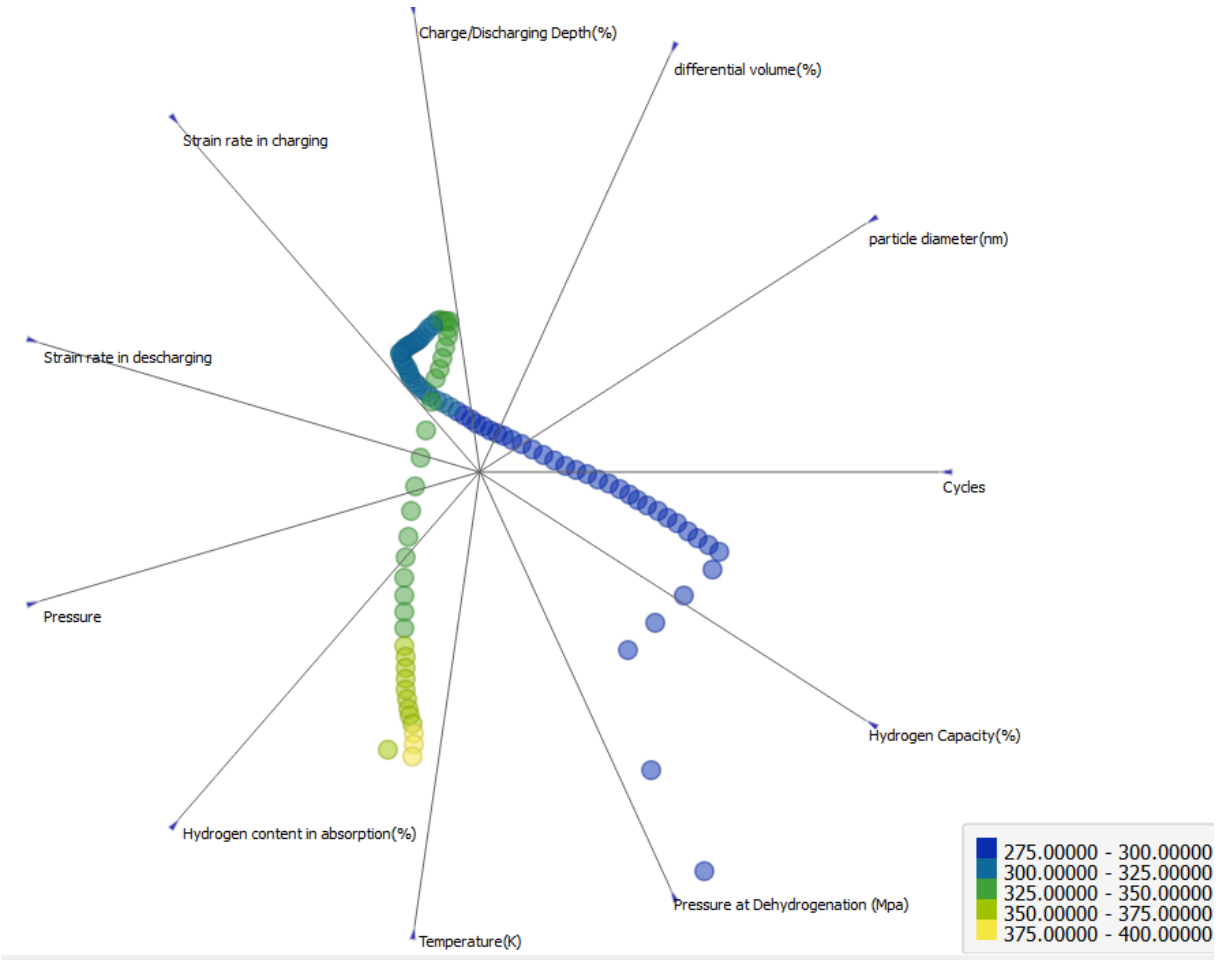

Figure 2: Free Viz Diagram

2.4 Data Preprocessing

Data preprocessing is a critical step in the machine learning pipeline, as it transforms raw data into a format that is amenable to machine learning algorithms. In this process, database where the dependent variables, typically located in the last column of the dataset, are labelled as 'y'. Concurrently, the features or independent variables are labelled as 'x'. This labelling is accomplished using a function known as LabelEncoder, a predefined function from the scikit-learn library. Following the labelling process, the data is then partitioned into training and test sets using a method known as random sampling. This method is a statistical technique used to select, define, and construct a representative subset from a large population. In this context, it is used to allocate 80% of the data for training the machine learning model, while the remaining 20% is reserved for testing the model's performance. This division ensures that the model is not only able to learn from a substantial portion of the data but also has an adequate amount of unseen data for evaluation.

2.4.1Feature Scaling

Feature scaling, was executed on both the test and training sets to ensure that no single feature dominated others due to its scale. This process is essential as many machine learning algorithms are sensitive to the scale of the input features. In this particular study, the standardization procedure was employed. This procedure transforms all features to have a mean of zero and a standard deviation of one, effectively bringing all features onto the same scale. This transformation was achieved using the StandardScaler function, a component of the scikit-learn library. The application of feature scaling via standardization ensures that all features contribute equally to the final prediction.

2.5 Modelling

Various regression machine learning models were employed and benchmarked against each other and scored using the value of adjusted R^2 value. The benchmarked models are Support Vector Regression (SVR), Decision Tree Regression (DT), Random Forest Regression (RF) and k Nearest Neighbour (kNN) algorithms. Support Vector Regression[9] is a variant of Support Vector Machine with the objective to find a hyperplane in an n-dimensional space based on the number of features or independent variables.

Support Vector Regression uses an insensitive tube to minimize error shown in equation 1[9]. This tube around the regression line has a width of Epsilon, measured vertically. Points falling inside the tube are considered to be within acceptable error range. Points outside the tube can result in different outcomes as they are not considered in the model.

$$\frac{1}{2}||w||^2 + c\sum_{i=1}^{m}(\xi_i^* + \xi_i) \tag{1}$$

Where, ξ_i^* is slack variable below the tube.

ξ_i is slack variable above the tube.

For a non-linear dataset a kernel trick known as gaussian radial based function (rbf)[10] was employed in order to linearize the data is shown by equation 2[10]

$$k(x\rightarrow, i\rightarrow) = e^{\frac{|-||x\rightarrow - i\rightarrow||^2}{2\sigma^2}} \tag{2}$$

Decision Tree Regression[9] uses a tree-like structure to make decisions and generate predictions based on the input data. It works with scatter plot to create decision trees where algorithm splits scatter plot based on information entropy. Now, the algorithm determines optimal splits by adding value to the grouping of points and stops when a certain minimum information is reached or when less than five percent of the total points are in a leaf. New predictions are assigned based on the terminal leaf where the new points fall.

Random Forest Regression[9] is ensemble form of learning that combines the power of multiple decision trees to generate more accurate and robust predictions. By combining the outputs of many decision trees, Random Forest Regression reduces the risk of overfitting and improves the stability and reliability of your models. In this method, K data points from the training set are selected in order to build the decision trees and process is repeated to create multiple decision trees. Now these multiple regression decision trees are used to predict the value of y for a new data point and taking the average of the predictions from all the trees, improving prediction accuracy.

The K-NN[9] algorithm is used for classification and regression tasks. It works by finding the k closest data points to a new input and predicting its label based on the majority class or average value of its neighbours. New data point is recognized based on its proximity to existing data. In this algorithm, k number of neighbours were chosen based on Euclidean distance.

Scikit learn[11] (also known as SK learn) is a free software machine learning package for Python. It features various classification, regression and clustering algorithms , and is designed to interoperate with the Python numerical and scientific libraries NumPy and SciPy. It allows for easy implementation of data structures and reduces the effort required to write code from scratch. Teaching computers machine learning concepts with Scikit-Learn simplifies the complex process using statistics, mathematics, and programming

The Key components of a good machine learning library are representation involves hypothesis space and formal language for expressing models and evaluation includes utility function, loss function, and judging one model versus another which the Scikit-Learn library contains[11]. The optimization helps in finding the best classifier due to having evaluation function with more than one Optimum indicates a well-optimized model in scikit-learn and prerequisites for working with scikit-learn include Python programming environment, numpy, pandas, matplotlib, and SciPy libraries. Scikit-Learn can solve clustering, density estimation, and dimensionality reduction problems. Clustering[11] aims to discover groups of similar data within a dataset. Density estimation determines the distribution of data within input space, while dimensionality reduction projects data from high dimensional space to lower dimensions for visualization.

2.5.1 Building the model

Scikit Learn Library is used in order to build all the models in the research. For the SVR model, rbf function was employed to linearize the non-linear data set and SVR function was induced from the scikit learn library. For DT model building, the binary tree was induced where the minimum number instance on the leaves was set to 20 with the tree depth limited to 100. The subset or leaf containing less than 5 data was discouraged to split which is the stopping criterion for the dataset. Then the DecisionTreeRegressor was induced from sci-kit library for training the model. For the RF model building, the estimators or no of decision trees are set to 20 while the limiting condition is made to not split subset with data less than 5. Then, RandomForestRegressor function is induced for the model for training the model with training sets. For this model, the value of K is taken as 20 which are said neighbours which we want to evaluate. After this Euclidean geometry is used to calculate the distance between the predicted point and neighbours. To build the model, kNeighborRegression Function is induced through the sci-kit library and the model is trained using training sets.

2.5.2 Training the model and making predictions

After the successful creation of model, the feature scaled training sets made by random sampling are fitted in the model to train the model. Now the trained model is used in order to predict the values using the features of the feature scaled test sets. This predicted data set and the original targeted value of test sets are visualized and compared further for the evaluation

2.6 Evaluation
Adjusted R^2 value method was employed in order to evaluate the machine learning models. Adjusted R-squared is a statistical measure that compares the fit of a model to the average of all possible models. It is a modified version of R-squared that accounts for the number of predictors in the model. A higher adjusted R-squared indicates a better fit of the model to the data[12]. It also penalizes the addition of unnecessary variables to the model. Adjusted R-squared values can range from 0 to 1, with values closer to 1 indicating a better fit. As the model contains more independent variables, it affects the totals sum of squares in regression analysis. Adjusted r squared accounts for unnecessary variables and avoids artificially inflating R square. The governing equation in calculating adjusted R^2 value depends upon the R- squared value which is calculated as shown in equation 3[12].

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (3)$$

where, $SS_{res} = SUM(y_i - y_i^*)^\wedge 2$, represents the square of error made between regression line and the predicted dataset.

$SS_{tot} = SUM(y_i - y_{avg}^*)^\wedge 2,$ represents the square of the error made between the actual data and average data of the dataset.

Adjusted R^2 value method can be expressed mathematically by equation 4[12].

$$Adjusted\ R2\ value = 1 - [\{1 - R^2\}\frac{n-1}{n-k-1} \quad (4)$$

where, K is the number of independent variables

N is the sample size

Therefore, Adjusted R^2 value method ensures adding variables only brings substantial improvement to the model.

2. **Results**

3.1 Dataset Preparation
Dataset was collected from the findings of Liu et al[5]. Around one hundred and five raw data were collected from the origin pro software and with the data visualization, five outliers were identified and erased from the database such that 100 data were finalized as the final dataset to be used in the model. The dataset includes twelve columns with 11 being independent variables (features) and 1 being the target (dependent variable). The independent variables contain cycles, Particle diameter, differential volume, charging/discharging Depth, Strain rate at charging, Strain rate at discharging, Pressure at absorption, Hydrogen content at absorption, Temperature at both charging and discharging, Pressure at Desorption and Hydrogen content at desorption. The target is Discharge capacity after each cycle. A glimpse of the dataset is shown in table 2.

Table 2: Prepared Dataset for the Study

Cycles	particle diameter(nm)	differential volume (%)	Charge/Discharging Depth (%)	Strain rate in charging	Strain rate in discharging	Pressure	Hydrogen content in absorption (%)	Temperature(K)	Pressure at Dehydrogenation (Mpa)	Hydrogen Capacity in desorption(%)	Discharge Capacity(mAh/g)
0	0	0	0	0.10722	0.60097	0	0	303	0	0	359.8371
1	1	0.054646	1	0.129182	0.60156	0.00152	0.0081	303	0.00136	0.2	393.2477
2	2	0.223273	2	0.151144	0.60215	0.00152	0.016162	303	0.00136	0.214141	384.947
3	3	0.389256	3	0.173106	0.60274	0.00152	0.032323	303	0.00136	0.228283	378.0988
4	4	0.465976	4	0.195068	0.60333	0.00152	0.048485	303	0.00136	0.242424	372.4439
5	5	0.452133	5	0.217031	0.603919	0.002575	0.064646	303	0.00136	0.256566	365.9589

3.2 Model Building

After the successful prediction of the dataset, to see how much predicted dataset depicts the original dataset, a plot was made which is shown below in figures 3-6 for each benchmarked model.

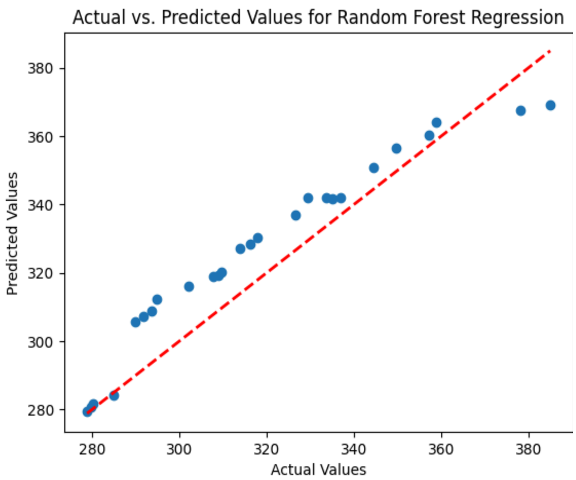

Figure 3 Actual vs Predicted Values for Random Forest Regression

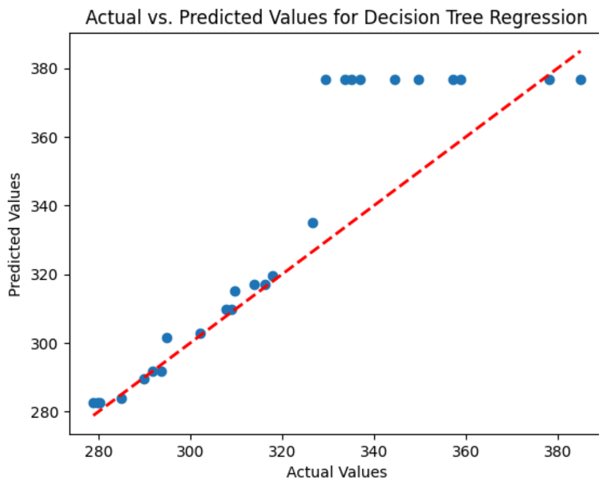

Figure 4 Actual vs Predicted Values for Decision Tree Regression

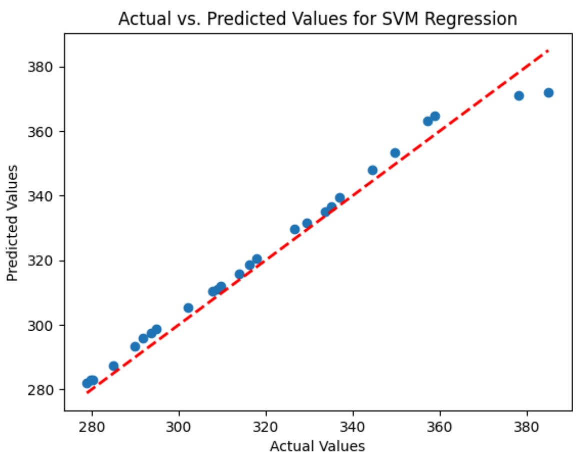

Figure 5 Actual vs Predicted Values for Support Vector Regression

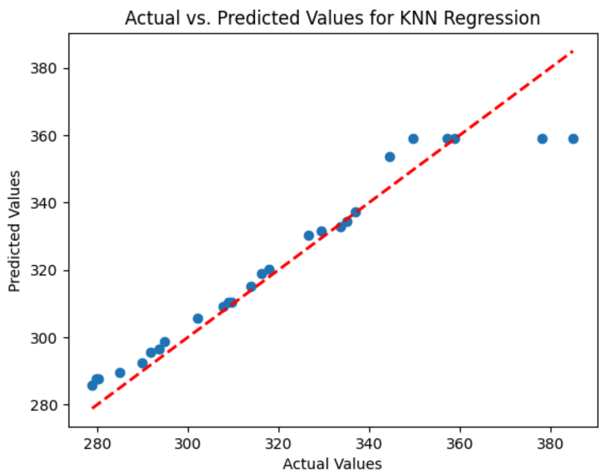

Figure 6 Actual vs Predicted Values for kNN Regression

3.3 Evaluation

Adjusted R-squared value for the different models is shown in figure 6 while the corresponding R-squared values are listed in figure 5. The SVR model performed best in both the evaluation values with the adjusted R-squared value of 0.9619 which is followed by kNN model which had the adjusted

R-squared value of 0.8824. The decision tree and random forest got affected by having more features and had a much lesser adjusted R-squared value compared to R-squared value.

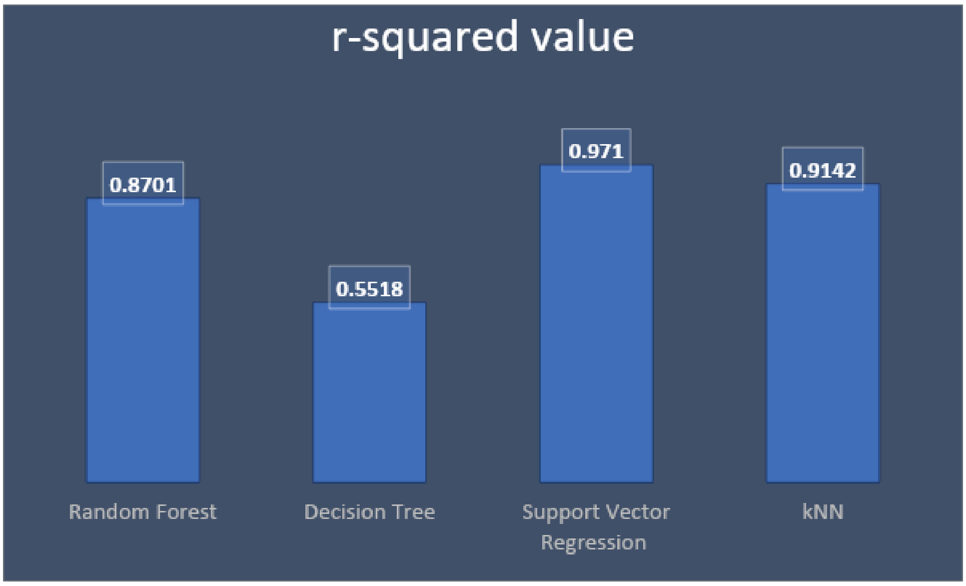

Figure 7 R- Squared Value of Models

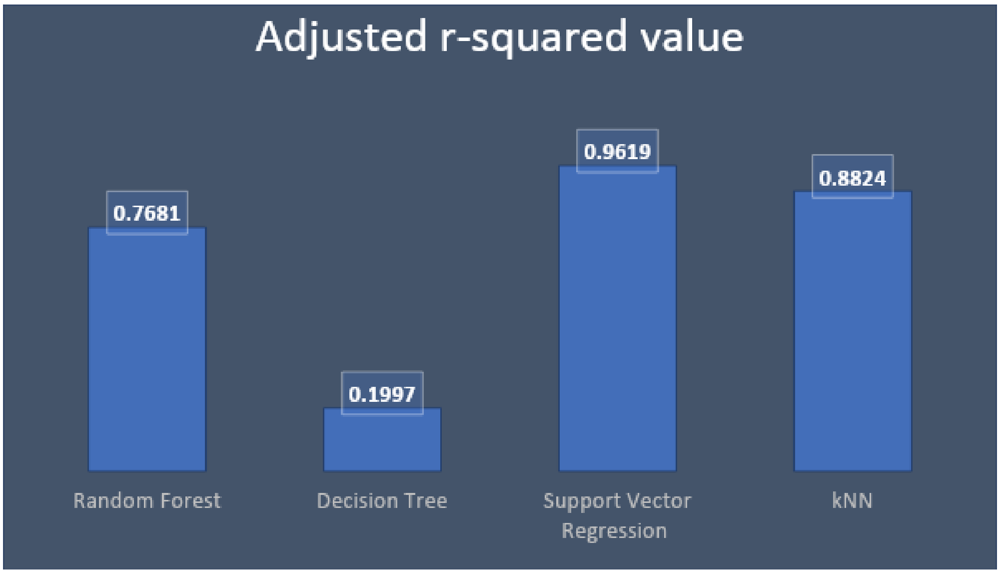

Figure 8 Adjusted R-squared Value of Models

3. **Discussion**

The study demonstrates the effective use of machine learning models to predict the performance of metal hydrides for hydrogen storage. The Support Vector Regression (SVR) model, in particular, showed high accuracy with an adjusted R-squared value of 0.9619. Among the four models tested, the K Nearest Neighbour (KNN) also performed well, indicating the potential of these models in optimizing hydrogen storage solutions. However, the Decision Tree and Random Forest models were less effective, possibly due to the complexity introduced by a larger number of features. A significant aspect of the research was the extraction of numerical data from graphical sources using Web Plot Digitizer, which underscores the importance of accurate data preparation in machine learning. The findings suggest that with further refinement, these machine learning techniques could be pivotal in advancing the field of hydrogen storage, particularly in enhancing the discharge capacity of metal hydrides. The methodology could also be extended to other materials and systems for hydrogen storage.

4. **Conclusion**

The study delves into the application of machine learning techniques to address the significant challenge of efficient hydrogen storage, focusing on the performance prediction of La-Ni-Mg Alloy, a metal hydride with promising attributes. Metal hydrides, particularly La-Ni-Mg Alloy, are highlighted for their safety, high storage capacities, and ease of handling.

The dataset, constructed from a combination of a secondary source and the Materials Project database, comprises of 11 features and a dependent variable (discharge capacity), representing various parameters influencing the alloy's discharge capacity after 100 cycles. The dataset used is obtained from the result of an experiment based on La– Mg-Ni-based alloys namely La_2MgNi_9, La_3MgNi_{14} and La_4MgNi_{19}. While the dataset represents only the features of La_2MgNi_9 alloy during the cycle. The dataset employed in this study was sourced from above mentioned experiment, thereby justifying its use in developing a reliable machine learning model.

The study utilizes four regression models - Support Vector Machine, Random Forest, Decision Tree, and K Nearest Neighbour (KNN) - showcasing the versatility of machine learning in predicting material performance. Dataset preparation involved overcoming the challenge of extracting numerical values from graphical representations, achieved through the use of Web Plot Digitizer. The Free Viz plot visually represents the relationships between independent variables and the target discharge capacity, offering insights into potential correlations and influences. Data preprocessing, a critical step, ensures appropriate labelling, division into training and test sets, and feature scaling to address algorithm sensitivity. The modelling phase employs four regression models trained on feature-scaled datasets. Evaluation metrics, specifically the Adjusted R-squared value, highlight the superior performance of the Support Vector Regression (SVR) model with an adjusted R-squared value of 0.9619, emphasizing its accuracy in predicting discharge capacity. The KNN model also exhibits commendable performance with an adjusted R-squared value of 0.88. However, the Decision Tree and Random Forest models display lower adjusted R-squared values, possibly influenced by an increased number of features. The study's implications extend to a deeper understanding of metal hydride performance and demonstrate the potential of machine learning in predicting material behaviour, with implications for optimizing hydrogen storage systems.

5. **Future recommendation**

The support vector Regression model has the highest R-squared value among the regression models used in this study. Thus, it can be used to predict the degradation of metal hydrides. Though the dataset used in this study uses only 100 cycles of charging and discharging of the metal hydrides. Machine Learning models require the use of an adequate amount of data to be fed into the model to ensure that the data can represent the underlying distribution well enough to obtain an unbiased and a reliable result. Thus, acquiring a larger dataset is necessary to further consolidate the reliability of the model in predicting the preformation of the metal hydrides.

6. **References**

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